Research Question:
Does within-lifetime reinforcement learning influence the rate at which effective food-seeking behaviours become encoded in heritable neural network weights across generations?

Experiment:
Environment:
A grid world with food dispersed. There are three food types, each of which carry a certain food score (energy value). For the sake of argument, the food types are red, orange and teal with scores of 0.5, 1 and 2 respectively. The world has the same structure, but there are landmarks indicating that a certain type of food is nearby, such as a teal wall indicating that teal food is nearby, as well as obstacles which will be obstructions. To survive, an agent must increase its energy by eating food. For each 10 ticks that go by, the agent loses energy; once the energy is 0 or the agent's lifetime is over, it dies. The food with higher scores are rare, so the agent needs to learn behaviours to eat these to survive for longer. At the end of a generation, we have two measures: lifetime reward (average of energy levels from all agents) and fitness (highest total food scores from the top 5 agents to account for stochastic favourable spawns). For a single run of learning vs. natural selection, the first agent spawns from the same location. This starting location changes between runs.

To establish the targets, we run pilot experiments in a different, simpler world to establish fitness and lifetime reward targets. This simpler world has a higher food abundance, fewer obstacles and more landmarks.This will be run for both the learning and natural selection conditions, choosing the highest outcome measures from the two. These measures will be used as the target values for the runs.

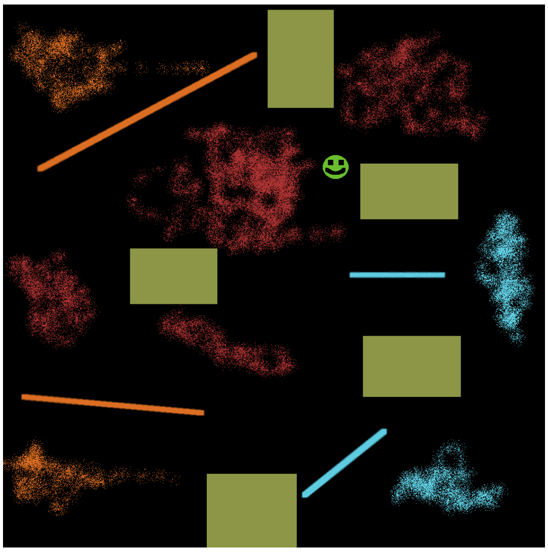

Figure 1: Hard Environment

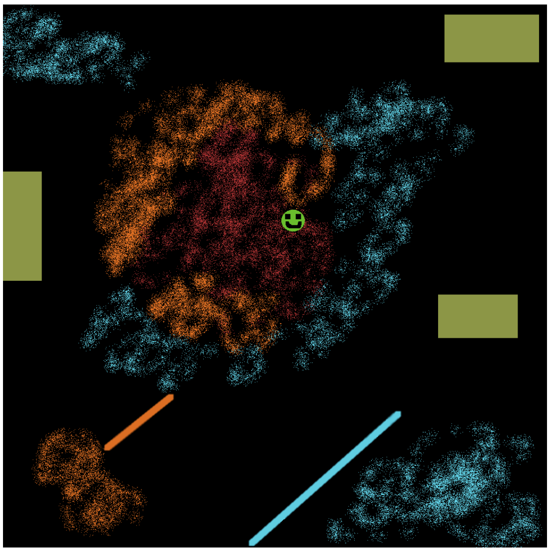

Figure 2: Simple Environment for pilot

In the given example environments, the coloured patches are food, the rectangles are obstacles, the blurred lines are landmarks and the smiley face is the starting location of the first agent.

Experimental conditions:

- Within-lifetime learning: the experiment is run where the agents learn through exploring the world. As they learn, the weights in the neural network change in response to decreasing energy levels and increasing energy from different food types, and this learning influences fitness. At the end of a generation, the top 5 highest fitness scores are taken, and these are used by the genetic algorithm to determine the initial weights of the brains of the next generation.
- Natural selection (no learning): the experiment is run and agents do not learn within their lifetime. Instead, only the genetic algorithm is used to determine the initial weights of the brains of the next generation.

Independent variable:

- Presence of within-lifetime reinforcement learning.

Dependent variables:

- Average lifetime reward: the average energy per generation will be used as it indicates how many agents do not die from energy levels that reach 0. The larger the number, the more successful the generation was at maintaining energy levels, which would be done by eating the food with higher scores more consistently. This will be done at the beginning of a generation (20% into its lifetime) to see if behaviour is becoming encoded and the end.
- Fitness: the fitness will be used as the score indicates how much food has been used. This metric is less robust than the lifetime reward as it only considers food and not whether the generation was able to continuously eat food. However, a higher value would indicate that more food was eaten or that more food with higher scores were eaten.

Hyper-parameters (many determined by pilot runs):

- Neural network architecture: inputs (food type detected in each eye direction, distance to nearest food, landmark signals detected, current energy level), outputs (movement in up, down, left, right directions). Input layer (8-12 neurons), hidden layer of (6-10 neurons) and output layer with 4 neurons.
- Lifetime of agent: provisional value of 1500 ticks. Currently, an agent's lifespan is 300 ticks which is quite short. Increasing this to 1500 allows enough time for the agent to learn. This might change based on the pilot runs, where we shall see when learning happens; if need be, this number will increase.
- Number of generations threshold per run: determined by the pilot run when learning converges. Provisionally, 1000 generations per run
- Number of runs: 20-30 runs for each experimental run. This will largely depend on how long a pilot run takes. Each run starts at a new spawn location
- Number of simulations: 4 (2 for establishing the baselines for learning and natural selection, and 2 for the experimental conditions for learning and natural selection)
- Mutation rate for the genetic algorithm: 0.01 to 0.05 for each weight
- Learning rate for within lifetime learning: will be determined and calibrated from the pilot runs. Learning rate must be low enough to allow for gradual convergence and not converge too quickly.